

The parabolic cylinder differential equation is of the form [40]:

$$\frac{d^2 y}{dz^2} + \left(n + \frac{1}{2} - \frac{1}{4} z^2 \right) y = 0 , \quad (4)$$

where n is an integer. Eq. (3) can be transformed into the form of Eq. (4) with the aid of the following substitution:

$$z = \sqrt{\frac{2\mu\omega}{\hbar}} x . \quad (5)$$

From Eq. (5) one finds that

$$\frac{d^2 \psi}{dx^2} = \frac{2\omega\mu}{\hbar} \frac{d^2 \psi}{dz^2} . \quad (6)$$

Using Eqs (6) and (5) in Eq. (3) leads to

$$\frac{d^2 \psi}{dz^2} + \left(\frac{E}{\hbar\omega} - \frac{z^2}{4} \right) \psi = 0 . \quad (7)$$

As we are expecting quantized energy levels we set $E = E_n$ and let

$$E_n = \hbar\omega(n + 1/2) . \quad (8)$$

With this substitution, Eq. (7) is in the form of the parabolic cylinder equation so that the state functions ψ_n are given by the complete orthogonal set of solutions for Eq. (7), the *parabolic cylinder functions*, $\{D_n(z)\}$ for $n = 0, 1, 2, \dots$. The state functions for the 1D oscillator are then:

$$\psi_n = N_n D_n \left(\sqrt{\frac{2\mu\omega}{\hbar}} x \right) , \quad (9)$$

where N_n is a normalization constant.

Not surprisingly, the functions, D_n can be written in terms of the Hermite polynomials, H_n as [40]

$$D_n(z) = \frac{1}{(\sqrt{2})^n} e^{-z^2/4} H_n \left(\frac{z}{\sqrt{2}} \right) , \quad (10)$$

which, due to the nature of the Hermite polynomials, restricts n to non-negative integers. Using the Rodrigues formula for the Hermite polynomials, [41], we can write an *effective Rodrigues formula* for the parabolic cylinder functions as

$$D_n(z) = (-1)^n e^{-\frac{z^2}{4}} \frac{d^n}{dz^n} e^{-\frac{z^2}{2}} . \quad (11)$$

The first six parabolic cylinder functions are listed in Table 1. A few of these functions are plotted in Figure 1.

The normalization constant can now be found using the following identity from Reference [42]:

$$\int_{-\infty}^{\infty} e^{-x^2} H_n^2(x) dx = 2^n n! \sqrt{\pi} . \quad (12)$$